

Weather Forecast Android Application Internship Project Documentation

Intern Information

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Domain: Android Development

Project: Weather Forecast Android Application

Introduction

This Android application provides real-time weather information using WeatherAPI. It is developed using Java and XML in Android Studio. Users can search weather by city or use their current location to view live weather updates, including hourly and 7-day forecasts.

Objective

To develop a modern Android Weather Forecast application demonstrating REST API integration, JSON parsing, RecyclerView, location services, Volley networking, Glide image loading, and Material Design UI.

Features

- Current Weather Information
- City Search
- Current Location Weather
- Live Temperature
- Weather Condition
- Humidity
- Wind Speed
- Rain Chance
- UV Index
- Hourly Forecast
- 7 Days Forecast
- Dynamic Weather Icons

Technologies Used

Java, XML, Android Studio, Volley Library, Glide Library, RecyclerView, WeatherAPI, Google Location Services, Git, GitHub.

System Workflow

Application Launch → Load Default City Weather → Fetch Live Weather Data → Parse JSON → Display Current Weather → Load Hourly Forecast → Load 7-Day Forecast → Display Weather Icons → Search City / Current Location → Update UI.

API Used

WeatherAPI Endpoint: <https://api.weatherapi.com/v1/forecast.json>

Future Enhancements

Air Quality Index (AQI), Sunrise & Sunset, Weather Notifications, Dark Mode, Favorite Cities, Offline Cache, Weather Maps, Multi-language Support.

Conclusion

The Weather Forecast Android Application demonstrates Android development using Java with live WeatherAPI integration, Volley, Glide, RecyclerView, JSON parsing, and location services. The project is suitable for internship submission.

Note

Screenshots are not included in this documentation. The complete source code is available in the GitHub repository and can be executed using Android Studio.